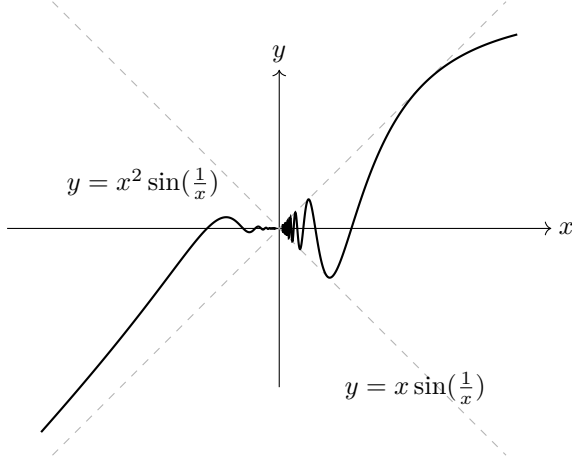


## 0.1 The function $x^a \sin(|x|^{-c})$

**Definition 0.1.1.** Define a map as: for  $a \in \mathbb{R}$  and  $c > 0$ ,

$$f_a^c : [0, 1] \rightarrow \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x = 0 \\ x^a \sin(x^{-c}) & \text{if } x \neq 0 \end{cases}$$



| Equivalent Condition Table | $f_a^c$        | $f'$           |
|----------------------------|----------------|----------------|
| Continuous                 | $a > 0$        | $a > 1 + c$    |
| Bounded                    | $a \geq 0$     | $a \geq 1 + c$ |
| Differentiable at $x = 0$  | $a > 1$        | -              |
| Bounded Variation          | $a \geq 1 + c$ | -              |

**Lemma 0.1.2.** The derivative of  $f_a^c(x)$  is given by:

$$\frac{d}{dx} f_a^c(x) = \begin{cases} \lim_{t \rightarrow 0} t^{a-1} \sin\left(\frac{1}{t^c}\right) & \text{if } x = 0 \text{ (provided the limit exists)} \\ ax^{a-1} \sin\left(\frac{1}{x^c}\right) - cx^{a-c-1} \cos\left(\frac{1}{x^c}\right) & \text{if } x \neq 0 \end{cases}$$

*Proof.* It is just simple calculation. Using this fact, we can establish the above table. □

## 0.2 Typical Examples

**Proposition 0.2.1.**  $f_1^1$  is continuous, **Not** differentiable at  $x = 0$ , **Not** of bounded variation, and does **Not** have a bounded derivative.

*Proof.* Since  $\left| x \sin \left( \frac{1}{x} \right) \right| \leq |x| \rightarrow 0$  as  $x \rightarrow 0$ ,  $f_1^1$  is clearly continuous.

Meanwhile,  $f'(0)$  exists if and only if  $\lim_{t \rightarrow 0} \frac{f(t) - f(0)}{t} = \lim_{t \rightarrow 0} \sin \left( \frac{1}{t} \right)$  exists.

However, this limit does not exist, so  $f_1^1$  is not differentiable at  $x = 0$ . To show that  $f_1^1$  is not of bounded variation, we construct a partition  $P_n$  as:

$$P_n \stackrel{\text{def}}{=} \left\{ 0, \frac{2}{n\pi}, \frac{2}{(n-1)\pi}, \dots, \frac{2}{2\pi}, \frac{2}{\pi}, 1 \right\}$$

Then, the variation satisfies:

$$V(f_1^1, P_{2n+1}) \geq \sum_{k=0}^{n-1} \frac{2}{(2k+1)\pi} = \frac{2}{\pi} \sum_{k=0}^{n-1} \frac{1}{2k+1} \rightarrow \infty \text{ as } n \rightarrow \infty.$$

Therefore,  $f_1^1$  is not of bounded variation. Similarly,  $f_n^n$  ( $n \in \mathbb{N}$ ) are not of bounded variation. □

But, in this way, we cannot determine whether  $f_2^1(x) = x^2 \sin \left( \frac{1}{x} \right)$  is bounded variation or not.

**Proposition 0.2.2.** If  $f : [a, b] \rightarrow \mathbb{R}$  is differentiable and the derivative  $f'$  is bounded, then  $f$  is of bounded variation.

*Proof.* For any partition  $P$  on  $[a, b]$ , by the Mean Value Theorem, there exists  $t_i \in (x_{i-1}, x_i)$  such that

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n \sup |f'| \cdot \Delta x_i = \sup |f'| \cdot 1$$

Since  $P$  was arbitrary,  $f$  is bounded variation. □

**Corollary 0.2.3.**  $f_2^1$  is of bounded variation on  $[-1, 1]$ .